Raku++

A from-scratch Raku, hand-written in C++17.

Interpreter + native compiler

runs in the **browser** via WebAssembly

Zero third-party dependencies

v0.9.1

NOT A RAKUDO FORK — SHARES NO CODE WITH IT

One implementation of the language, from lexer to native binary.

A hand-written lexer, parser, and tree-walking evaluator for real **Raku** — classes, roles, grammars, regexes, multi-dispatch, junctions, lazy sequences, a bignum tower, Unicode-correct strings, concurrency. It can also **compile** a program to a standalone native executable, and — as **Raku.js** — run client-side in a browser.

- **C++17, no deps**
  Hand-rolled bignum, Unicode tables, and regex engine. Builds anywhere CMake and a C++17 compiler run.
- **Interpret *or* compile**
  The same binary tree-walks a script or transpiles it to C++ and compiles native.
- **Measured, not claimed**
  Graded test-by-test against Roast, the official Raku specification suite.

GRADED AGAINST ROAST — THE OFFICIAL SUITE

How much Raku actually runs.

88.2%

189,102 / 214,384 declared tests pass — 97.4% of the tests that actually run.

558 / 1,462

files pass on the strict **all-or-nothing** bar — every assertion in the file must pass (~38%).

~3½min

to run the whole suite through a **self-hosted harness** — written in Raku, run by Raku++ itself.

Every release is gated on the full suite with **zero fully-passing-file regressions**.
Numbers are measured on each tag, never projected.

THE BREADTH OF WHAT'S IMPLEMENTED

Big-language Raku, most corners lit.

NUMBERS
Exact by default
Arbitrary-precision `Int`, exact `Rat` — `0.1+0.2-0.3` is exactly `0` — `Num`, `Complex`, allomorphs.

OBJECTS
class · role · grammar
Multi-dispatch, `BUILD` / `TWEAK`, mixins, `augment`, full `.^` introspection, `.wrap`.

REGEX
Grammars that parse
Backtracking engine, Unicode classes, `token` / `rule`, `.parse` with action classes & `make`.

SIGNATURES
multi & proto
Type / literal / `where` constraints, destructuring, coercion types, `callsame`.

FUNCTIONAL
Lazy & higher-order
Closures, `*` WhateverCode, junctions, `gather` / `take`, feeds, `1,2,*+*...Inf`.

METAPROG
Your own operators
User ops in all six categories with working precedence traits and meta-operators over them.

CONCURRENCY
Real OS threads
`start` / `await`, `Supply` / `react` / `whenever`, `Channel`, atomics; opt-in true parallelism.

UNICODE
Grapheme-correct
UAX #29 (emoji ZWJ), NFC/NFD/NFKC/NFKD, UCA collation — generated UCD/UCA 17.0 tables.

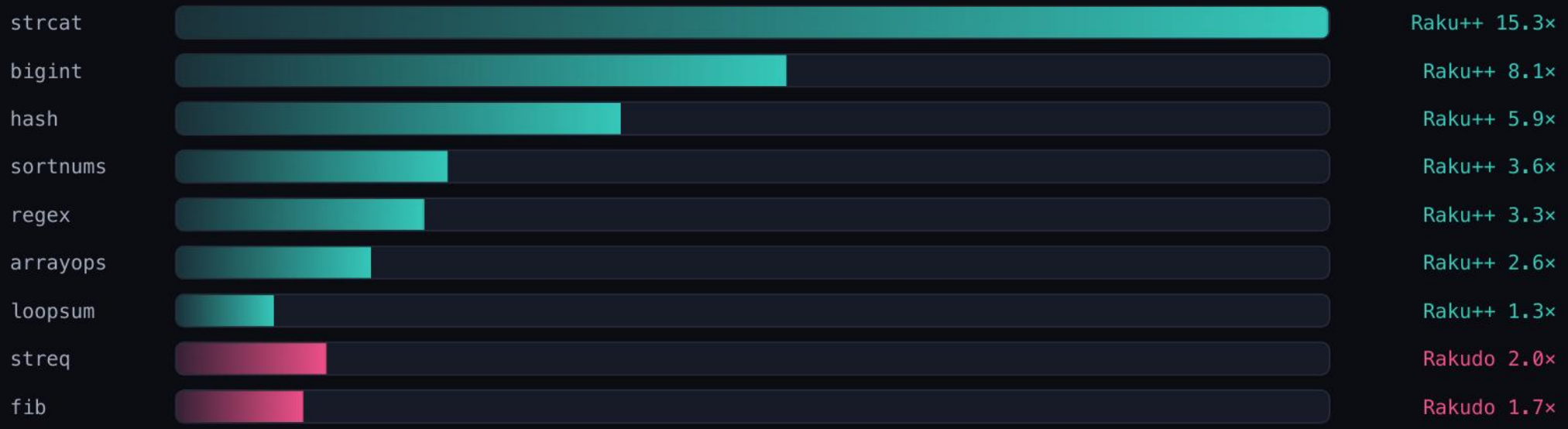
ONE SOURCE, FIVE TARGETS

Four ways to run — and a fifth in the browser.

`rakupp prog.raku`	**Interpret** — the default; covers the whole language.
`--bundle -o prog`	Standalone binary that re-parses the source at startup.
`--aot -o prog`	Standalone binary embedding the already-parsed AST.
`--exe -o prog`	**Transpile to C++ and compile native** — anything unsupported falls back to `--bundle` automatically.
`Raku.js → WASM`	**In the browser** — same semantics as native, client-side, no server, with an embeddable playground.

INTERPRETER · × VS RAKUDO V2026.06 · BYTE-IDENTICAL OUTPUT GATE

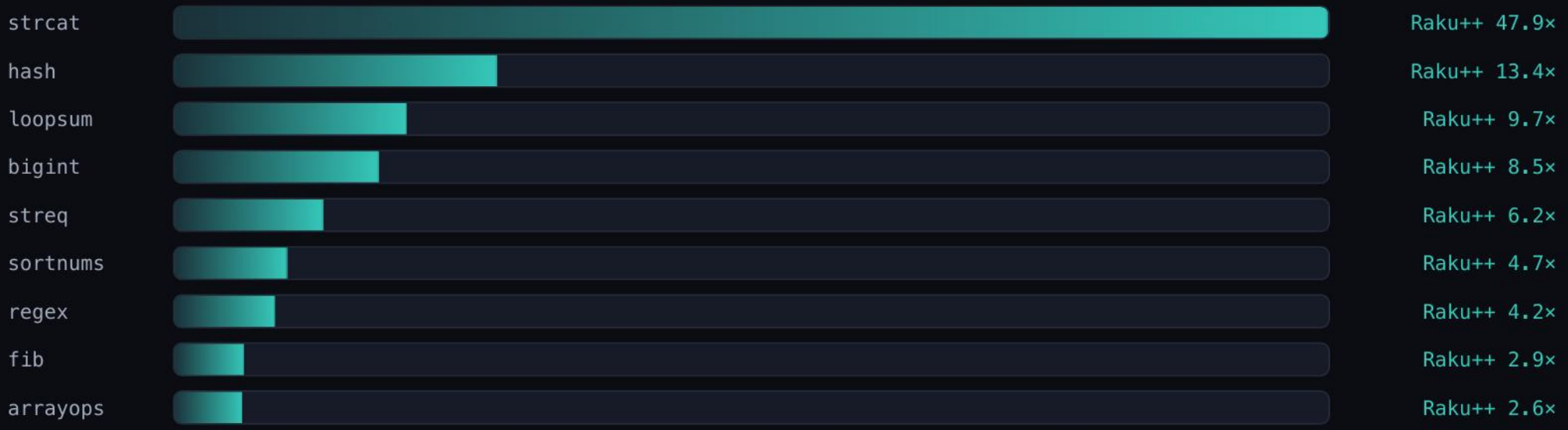
Fast to start, fast to run.



~2 ms cold start (best 1.8 ms of a 200-spawn loop) — a tiny native binary, no VM to spin up. Even interpreted, Raku++ wins 7 of 9 kernels; the two it trails on — streq, fib — the next slide compiles away.

RAKUPP --EXE · TRANSPILED TO C++, COMPILED NATIVE · × VS RAKUDO

Compiled, it wins every kernel.



--exe strips the tree-walk: every benchmark lands 2.6× to 47.9× ahead of Rakudo — streq and fib, the interpreter's only two losses, now included. Anything the compiler can't lower falls back to bundling automatically.

ONE INTERPRETER BEHIND EVERY SURFACE

A small constellation, one source of truth.

Everything downstream ships the *same* interpreter — the wasm, the two sites, the Homebrew tap, and the release binaries all trace back to `src/`. The spec inherits a new engine for free the moment the playground redeploys.

— MID-SIZE PROGRAMS, EACH LEANING ON A DIFFERENT PART OF THE LANGUAGE

Real programs, written in Raku.

Scheme
A Lisp interpreter on a Raku grammar.
GRAMMAR · EVAL

Forth
Stack machine, also grammar-parsed.
GRAMMAR

Markdown → HTML
Converter — also runs in the browser.
REGEX · WASM

JSON
Parser + formatter, browser-ready too.
GRAMMAR · WASM

Pastebin
HTTP service on raw sockets.
SOCKETS

Chat server
Concurrent, many-client.
THREADS

Key-value store
Its own wire protocol.
SOCKETS

rakus
A static-file HTTP server.
SOCKETS

Plus larger experiments — a **JavaScript/TypeScript** interpreter, a **Perl 5** interpreter (Raku parsing its own ancestor), and a **raytracer**. Every one of the 24 bundled `examples/` also compiles natively with `--exe`, byte-for-byte identical.

BEYOND WHAT ROAST'S UNIT GRANULARITY EXERCISES

Real applications run unmodified.

RAKU-COURSE

~1,500 pages, byte-identical

The full course static-site generator regenerates the entire site byte-for-byte identically to Rakudo — including the real zef-installed `YAMLish` module.

COVID.OBSERVER

End-to-end build

The site builder runs start to finish on Raku++ — a genuine data-processing application, not a curated snippet.

```
# exact arithmetic, lazy infinite sequence, grapheme strings —
# the things that make real programs match, not just tests
say 0.1 + 0.2 - 0.3;              # 0  (exact Rat)
say (1, 2, *+* ... *)[^10];  # 1 1 2 3 5 8 13 21 34 55
say "café".chars;               # 4  (graphemes, not bytes)
```

HONEST ABOUT THE EDGES

Not there yet.

- **Macros / RakuAST / slangs** — compile-time metaprogramming and pluggable syntax.
- **libffi-grade NativeCall** — structs and callbacks (basic C FFI to libc / <math.h> already works, without libffi).
- Some **IO** and **POD** corners.
- **Lock-free parallel atomics.**

Early-stage and growing test-first — the direction is more of Roast, passing file by file, with every release gated on no regressions.

MACOS · LINUX · WINDOWS · THE BROWSER

Try Raku++ today.

```
$ brew tap ash/rakupp   # macOS
```

```
$ brew install rakupp
```

```
↗ raku.online   # run it in the browser, no install
```

github.com/ash/rakupp · raku.online · spec.raku.online

Prebuilt archives on every GitHub Release — macOS universal, Linux static, Windows static-CRT — plus the WebAssembly bundle. No runtime dependencies, anywhere.